

Framework Selection Justification

Business process chosen: a document question-answering assistant, a user uploads a PDF and asks natural-language questions about its contents, and the agent answers using only that document. This is genuinely suited to agent automation: it's multi-step (read the document, find the relevant part, produce an answer), it needs to work over arbitrary external content supplied at runtime rather than a fixed knowledge base, and it currently requires a human to read the whole document themselves.

I compared three frameworks before writing any code. AutoGen structures work as a back-and-forth conversation between agents until they converge on an answer, which suits open-ended reasoning tasks but is a poor fit here: a question-answering pipeline doesn't need multiple agents negotiating with each other, and AutoGen's conversational loop makes it harder to guarantee the process terminates in a fixed, predictable number of steps. CrewAI is a role-based abstraction on top of LangChain — you assign agents roles and goals — which is fast to prototype with, but it's built around orchestrating multiple collaborating agents, which is unnecessary overhead for a single linear pipeline with no need for role delegation.

LangGraph was the right fit because this task is fundamentally a deterministic three-step pipeline — load the document, retrieve the relevant passages, produce an answer — with no need for open-ended negotiation between agents. LangGraph represents this as an explicit state graph, so every step's input and output is typed and inspectable, which made it straightforward to add the ReAct-style Thought/Action/Observation trace capture this assignment requires, and to later extend the pipeline (e.g. adding a real LLM generation step) without restructuring the whole thing. The trade-off is that LangGraph has more boilerplate than CrewAI for genuinely multi-agent tasks, but for a single-agent, single-purpose pipeline like this one, that overhead doesn't apply — LangGraph's explicitness was a net benefit, not a cost.